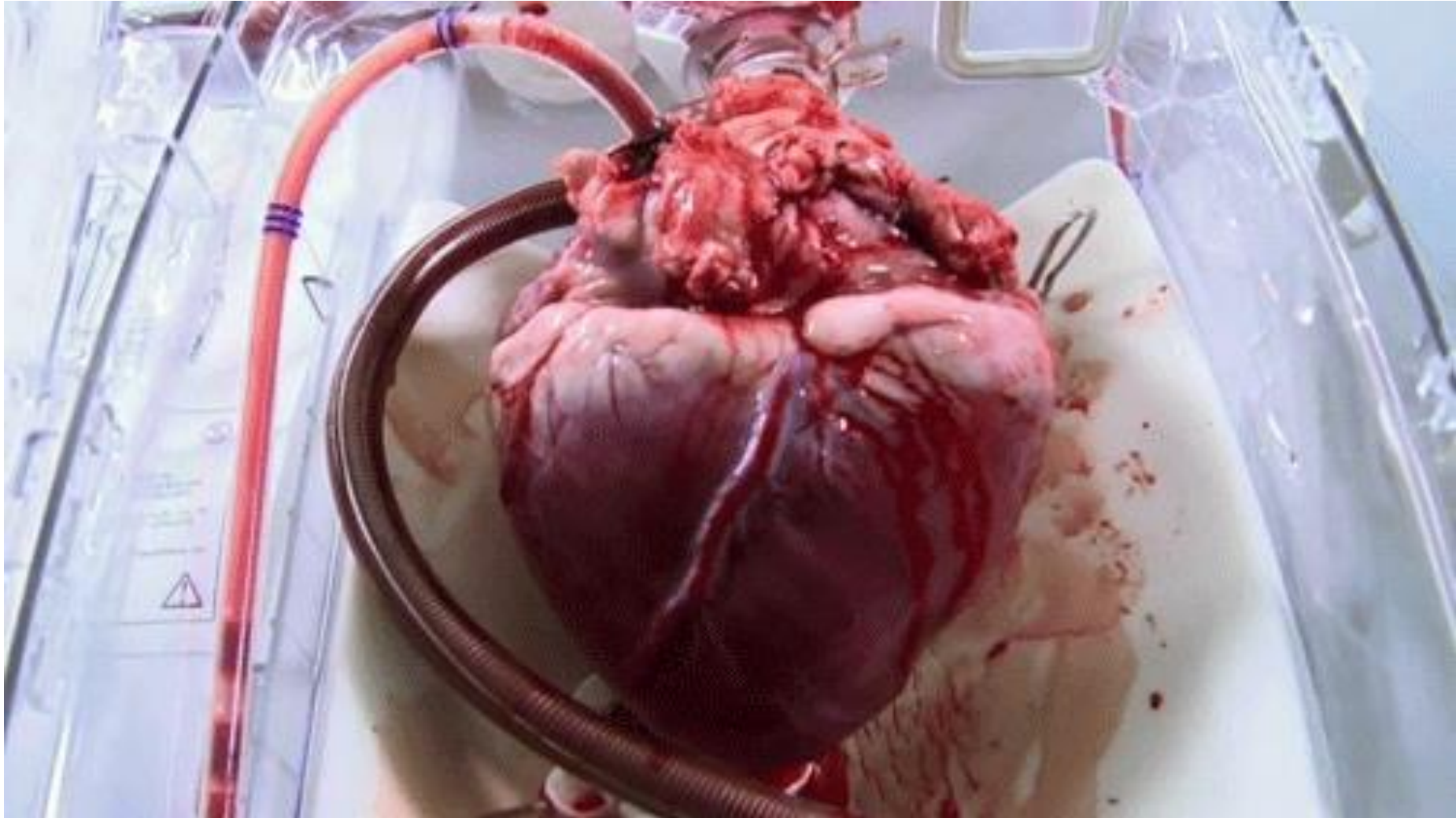
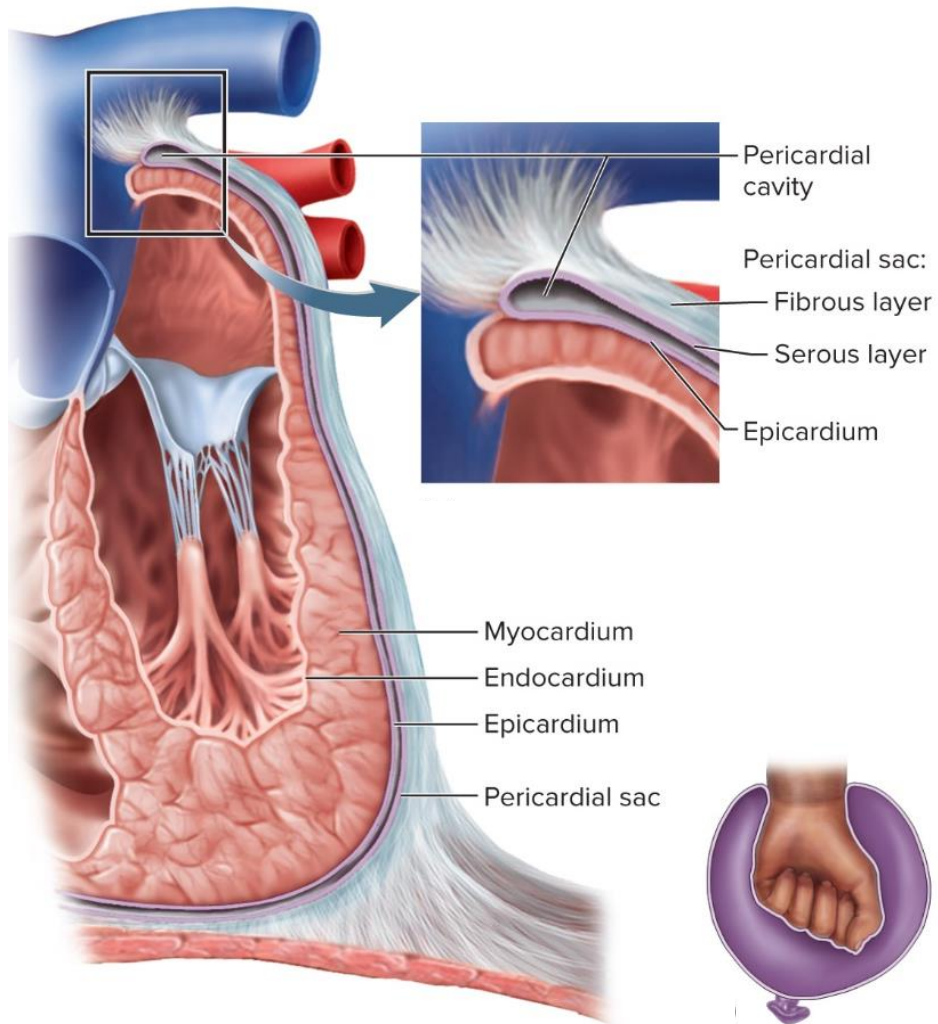


Exercise 14: The Heart

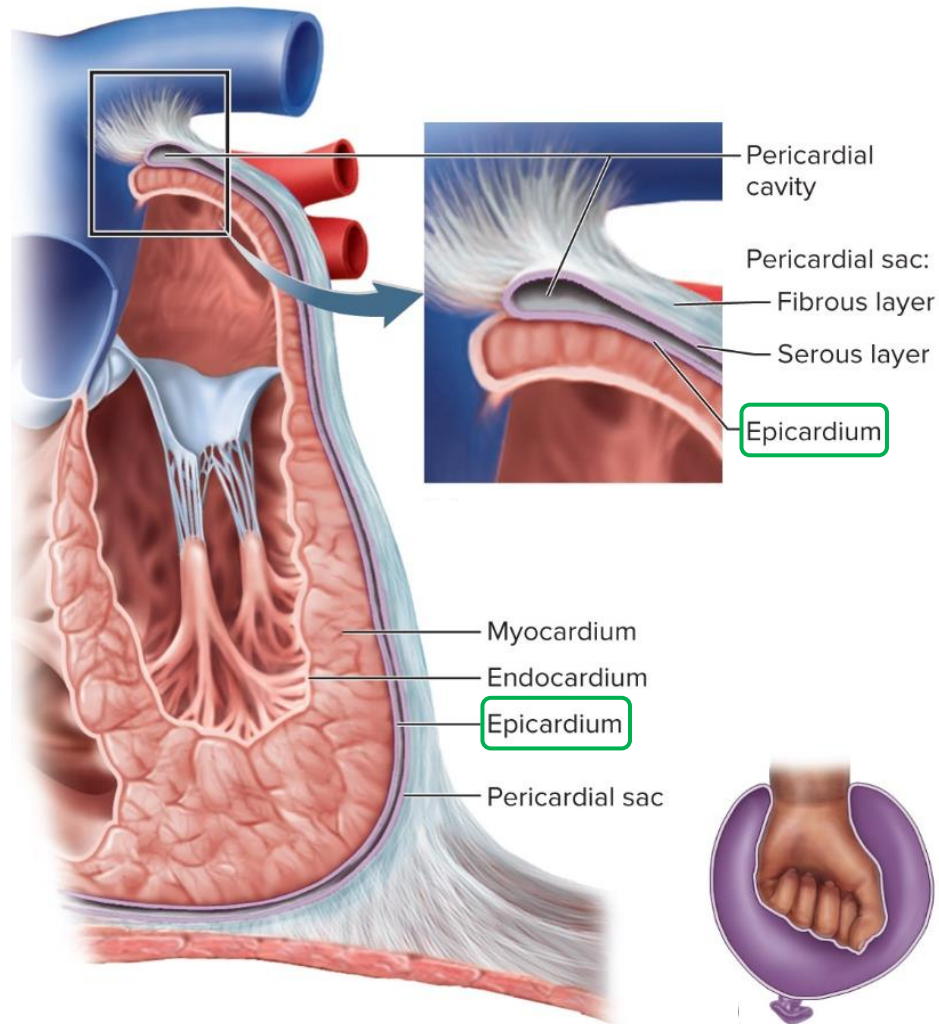


Cardiac Anatomy

*the heart wall consists of three layers



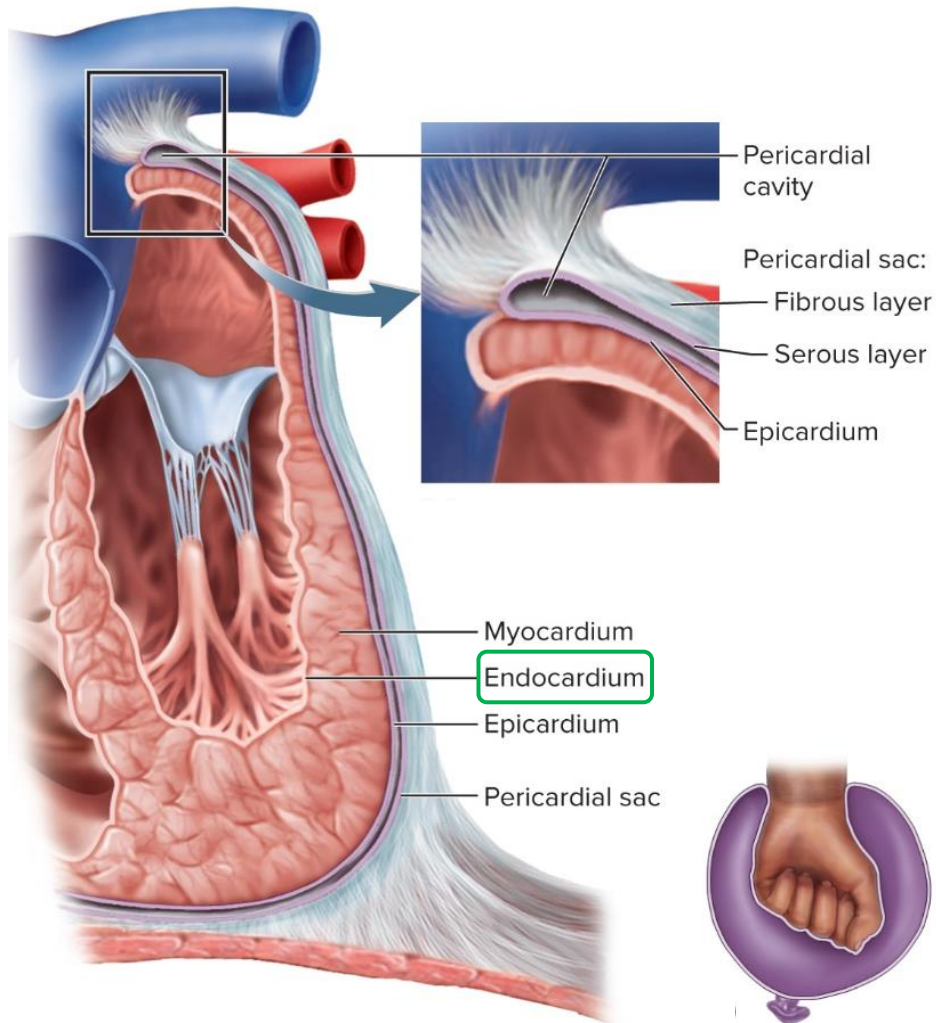
Cardiac Anatomy



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****epicardium*** (visceral layer of the serous pericardium) is a serous membrane of the external heart surface

Cardiac Anatomy

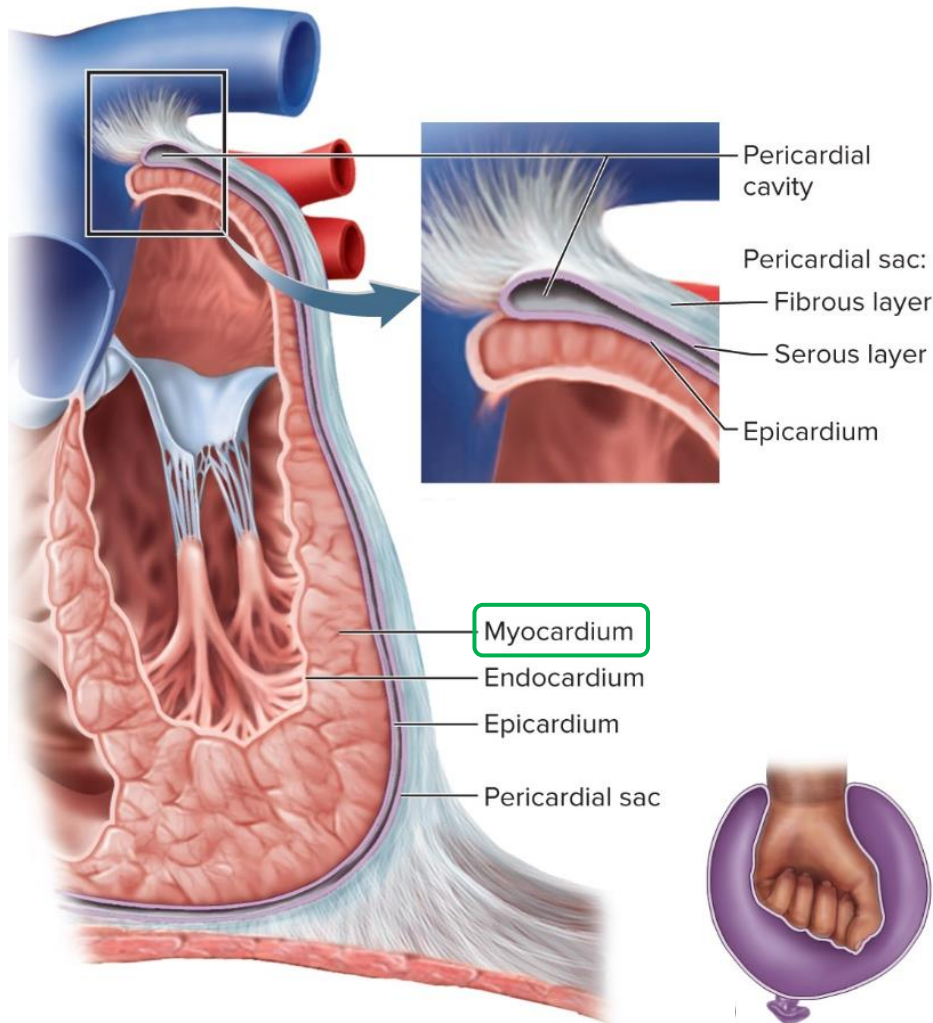


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****epicardium*** (visceral layer of the serous pericardium) is a serous membrane of the external heart surface

****endocardium*** lines the interior of the heart chambers, covers the valve surfaces, and is continuous with the endothelium of the blood vessels

Cardiac Anatomy



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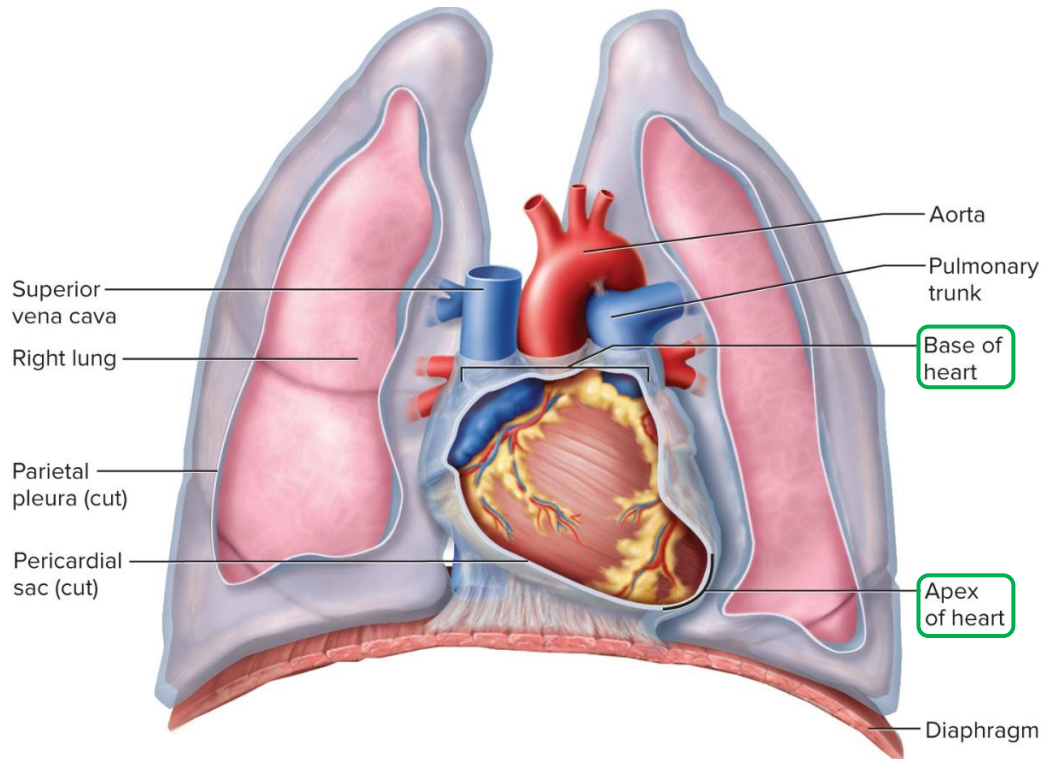
****endocardium*** lines the interior of the heart chambers, covers the valve surfaces, and is continuous with the endothelium of the blood vessels

****myocardium*** is composed of cardiac muscle

Cardiac Anatomy

***base** of the heart is superior where the great vessels are attached

***apex** of the heart is at the inferior end, just above the diaphragm

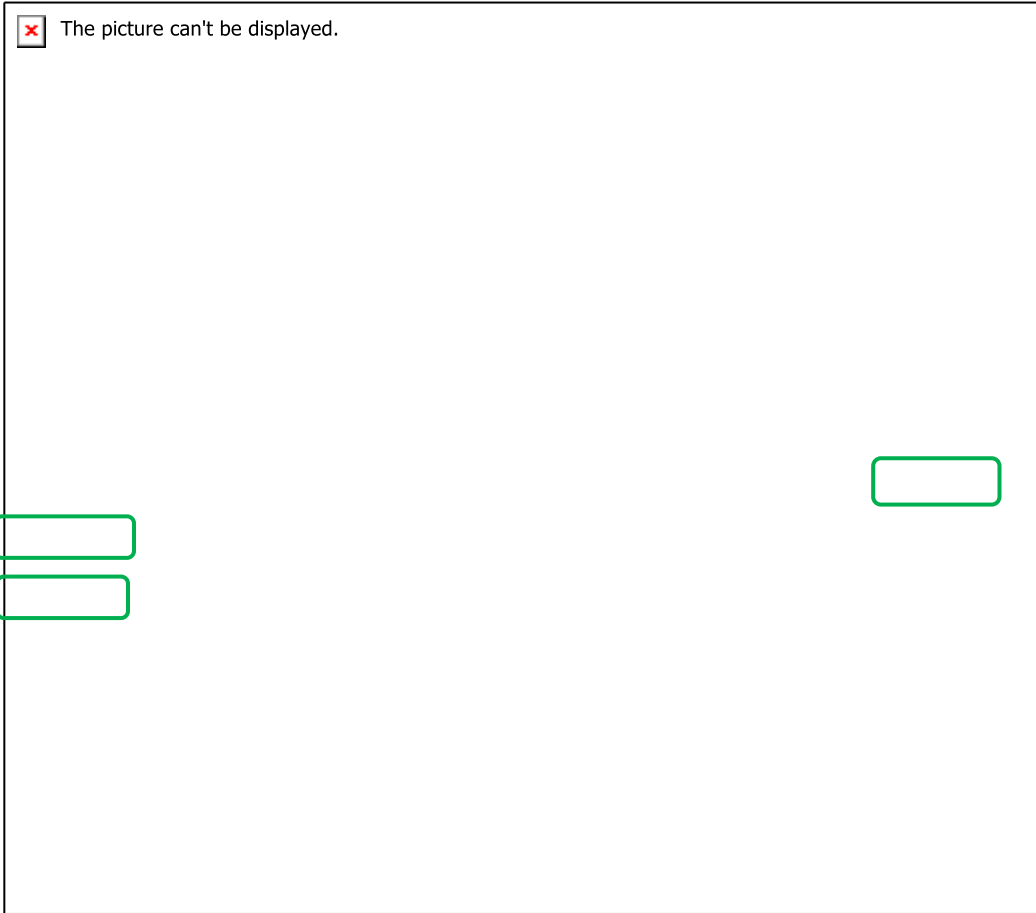


Cardiac Anatomy

*heart has four chambers

*two superior chambers are the ***right*** and ***left***
atria (receive blood returning to the heart)

*each atrium has an earlike flap called an
auricle that slightly increases volume

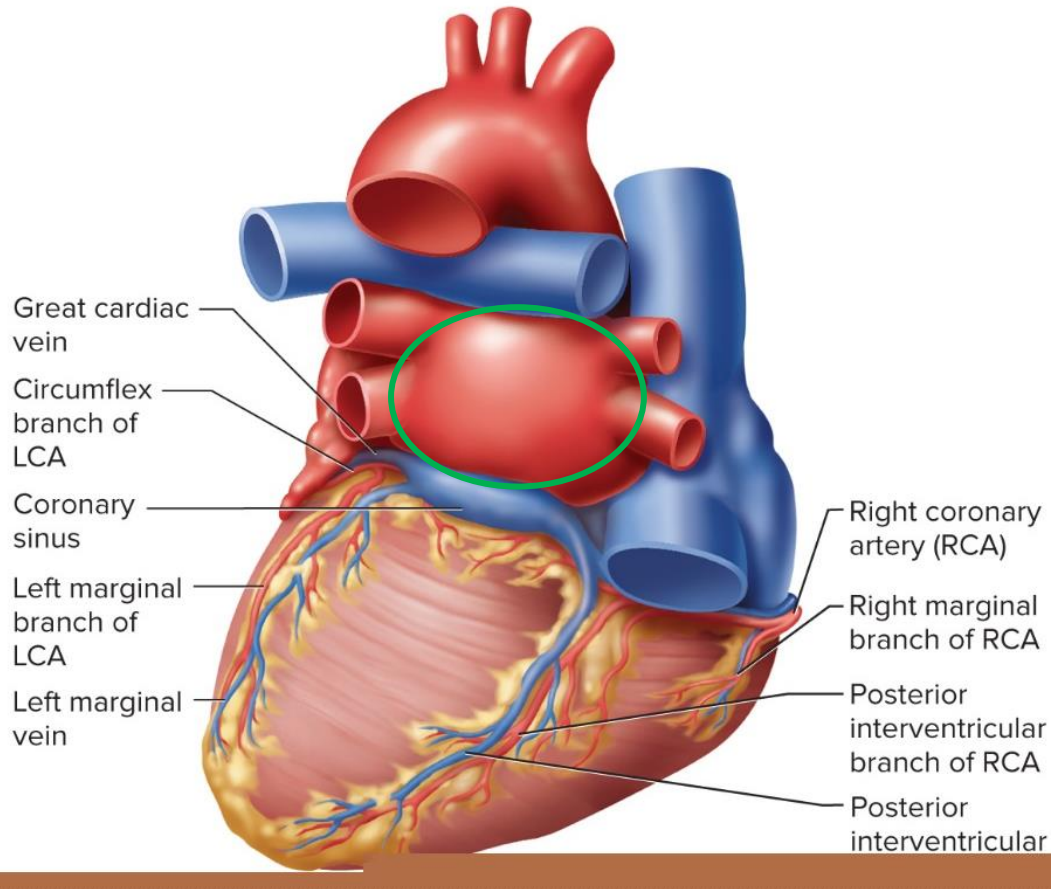


Cardiac Anatomy

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Cardiac Anatomy

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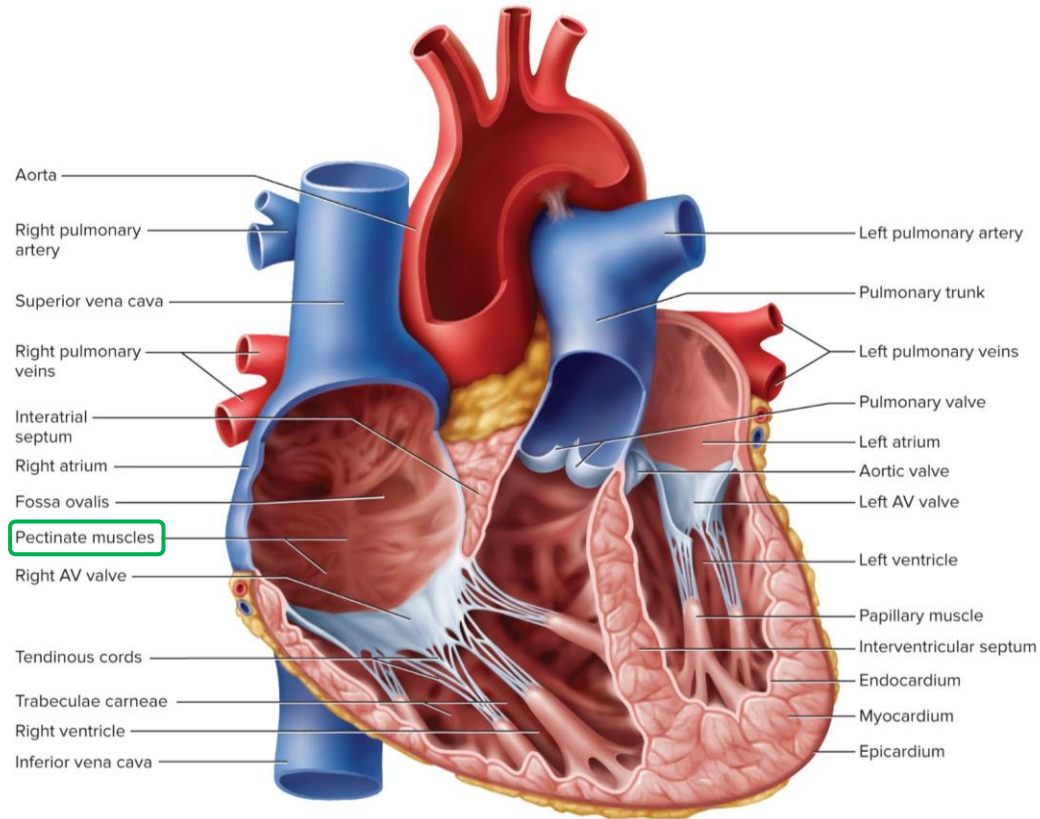
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- ****ligamentum arteriosum*** is a remnant of the ductus arteriosus

Cardiac Anatomy



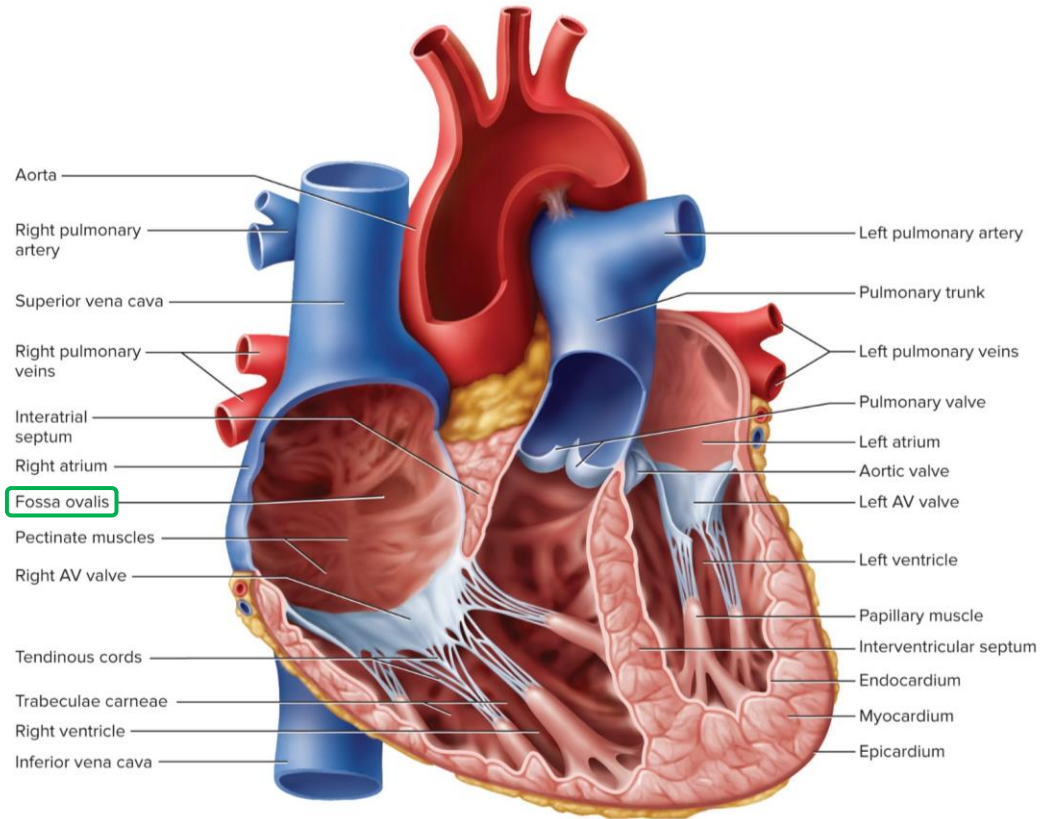
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Cardiac Anatomy



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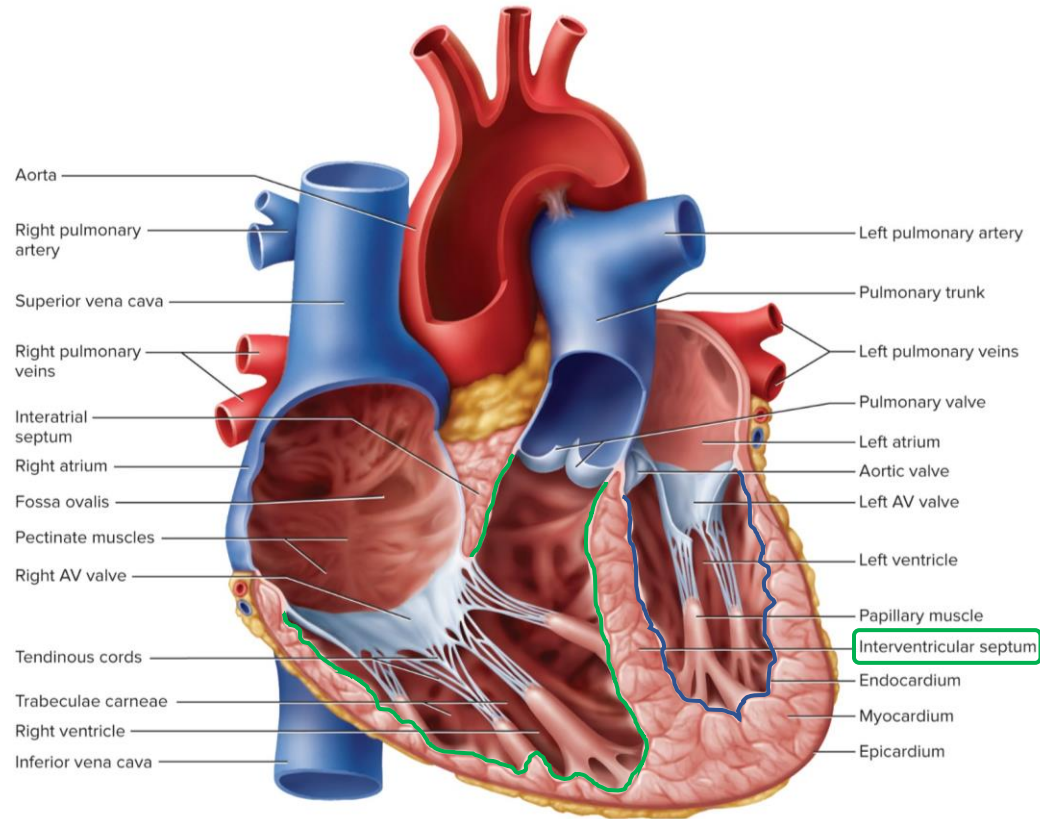
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***fossa ovalis** is a remnant of the foramen ovale

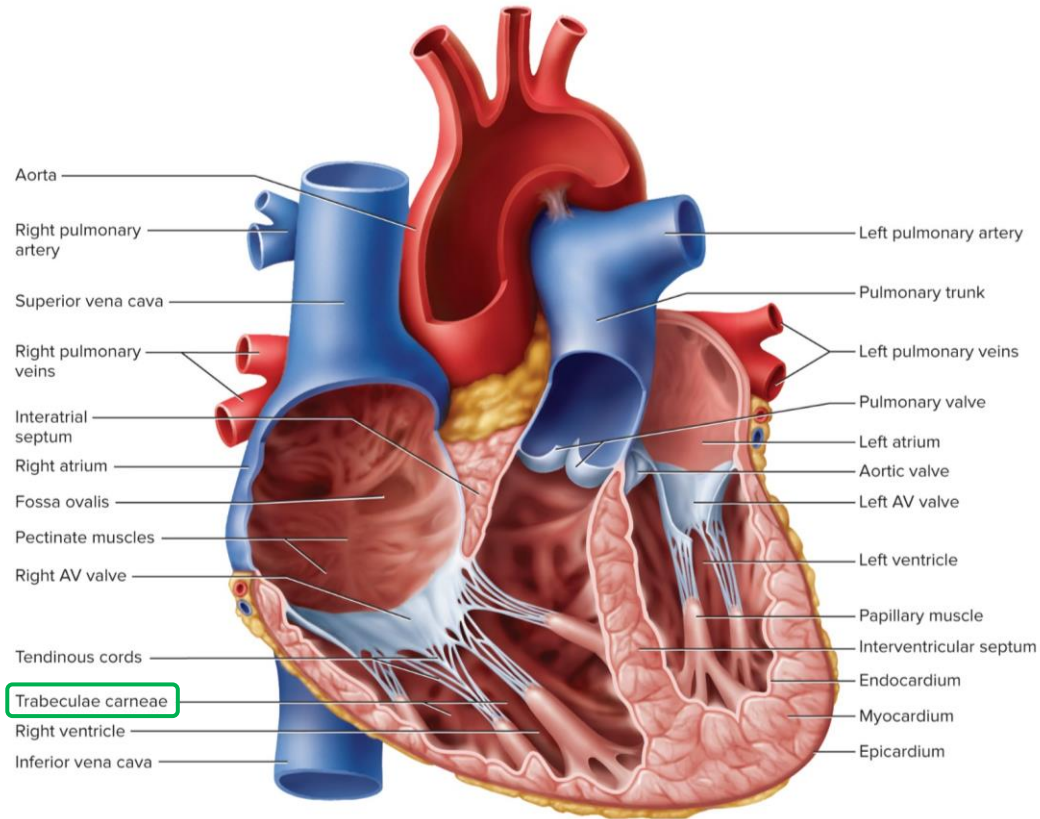
Cardiac Anatomy



*two inferior chambers, the ***right*** and ***left ventricles***, are pumps that eject blood into the arteries

*ventricles are separated by a thick muscular wall called the ***interventricular septum***

Cardiac Anatomy

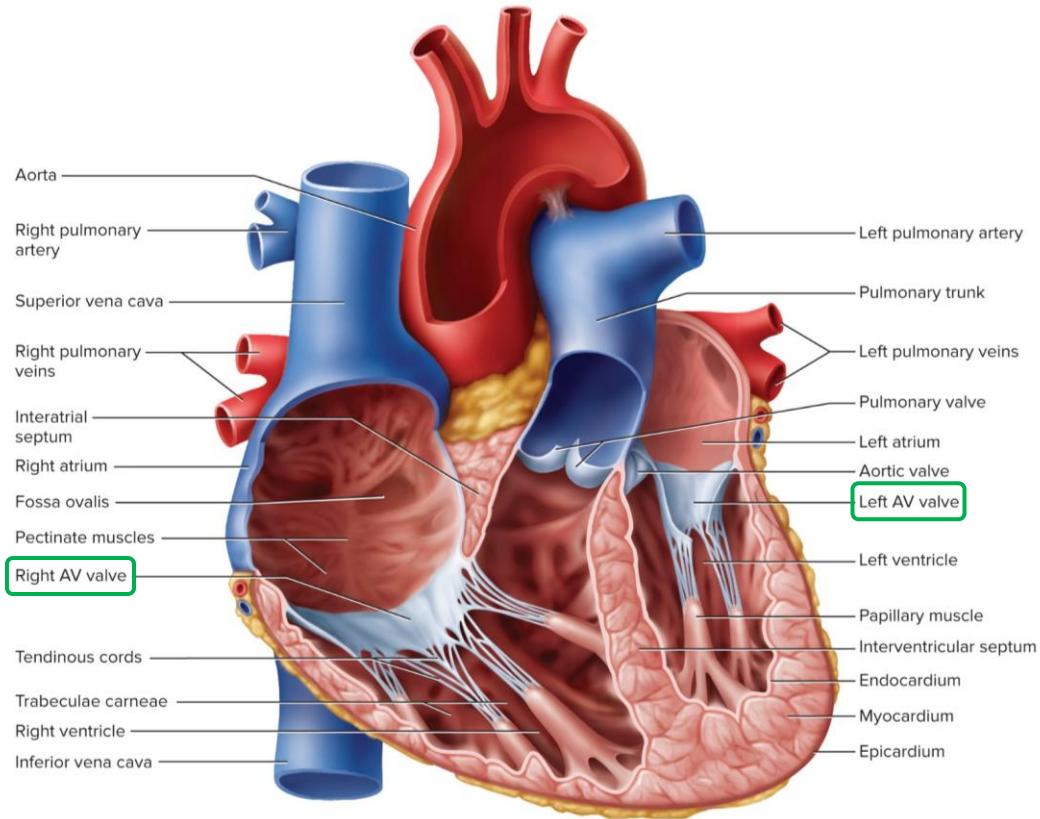


*two inferior chambers, the ***right*** and ***left ventricles***, are pumps that eject blood into the arteries

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*both ventricles exhibit internal ridges called ***trabeculae carneae***

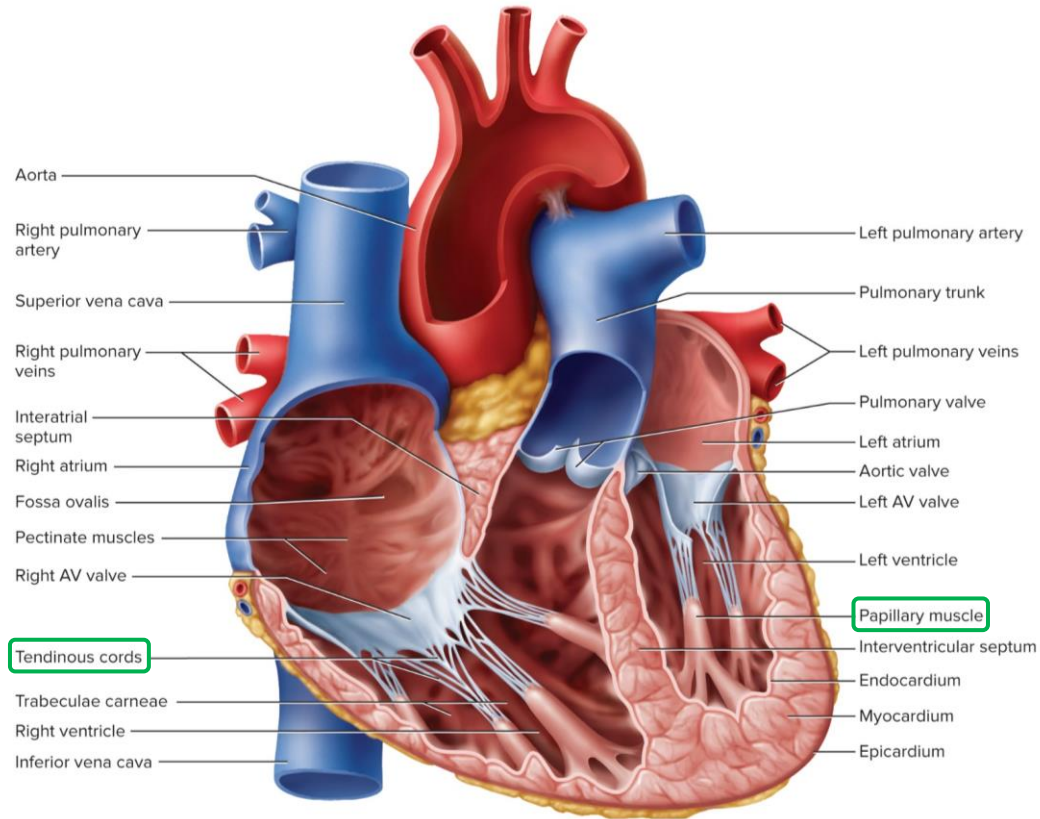
Cardiac Anatomy



*the **atrioventricular (AV) valves** regulate the openings between the atria and ventricles

***right AV (tricuspid)** valve has three cusps and the **left AV (mitral)** has two

Cardiac Anatomy

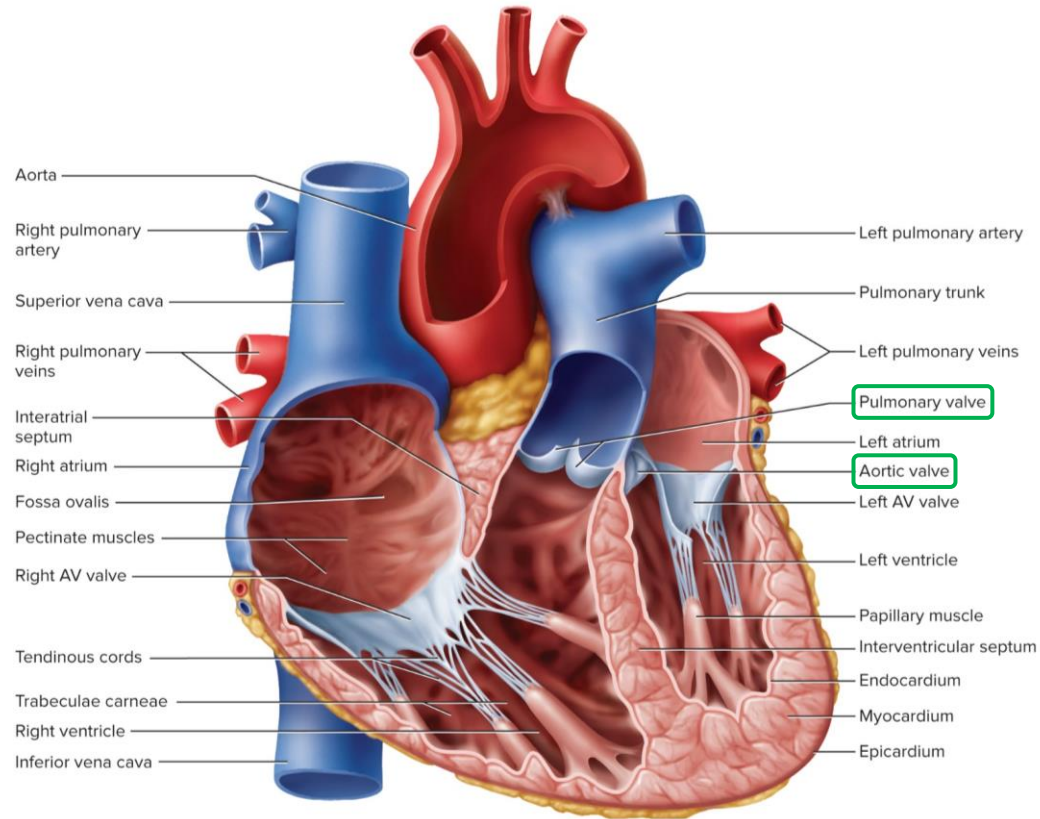


*the **atrioventricular (AV) valves** regulate the openings between the atria and ventricles

***right AV (tricuspid)** valve has three cusps and the **left AV (mitral)** has two

***tendinous cords (chordae tendineae)** connect the valve cusps to **papillary muscles** and prevent valves from flipping inside out

Cardiac Anatomy

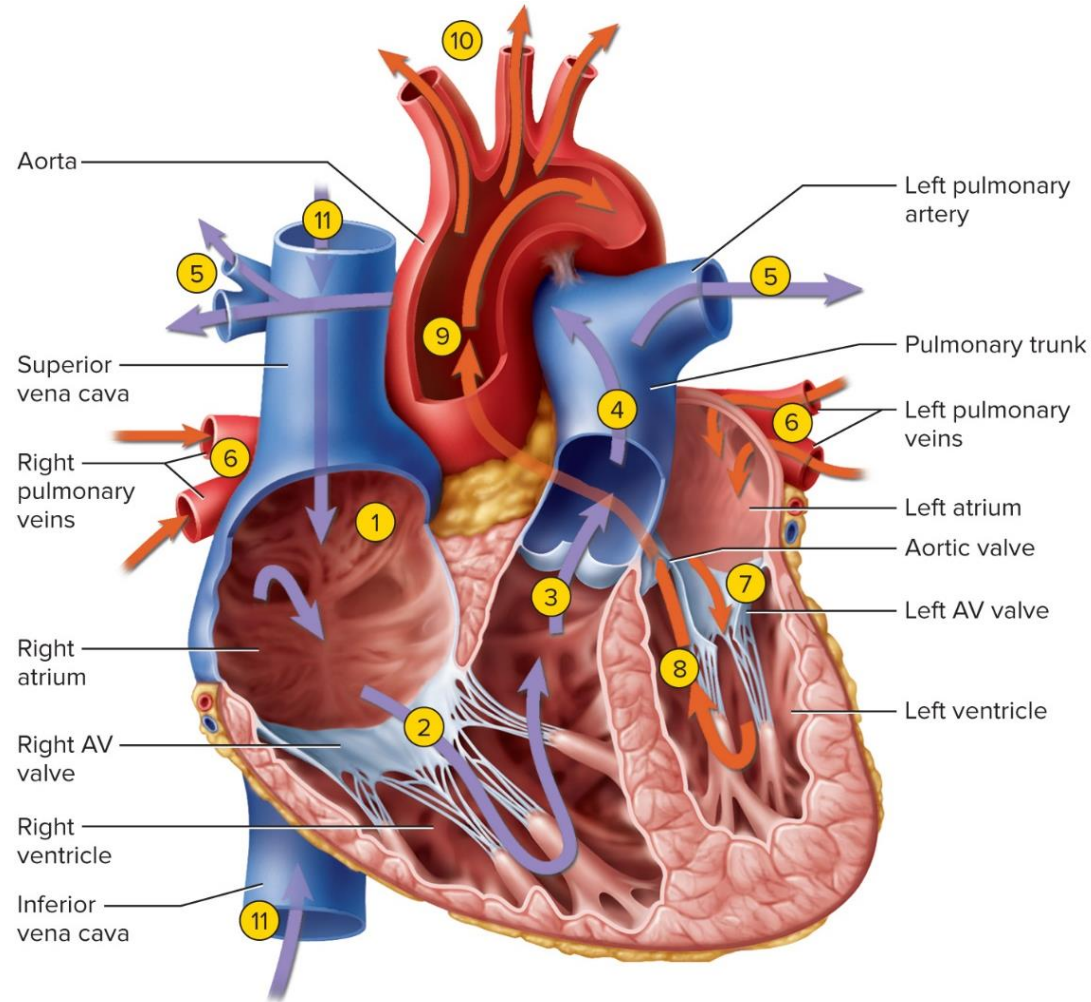


*the ***semilunar valves*** regulate the flow of blood from the ventricles into the arteries

****pulmonary valve*** controls the opening from the right ventricle into the pulmonary trunk

****aortic valve*** controls the opening from the left ventricle into the aorta

Cardiac Anatomy

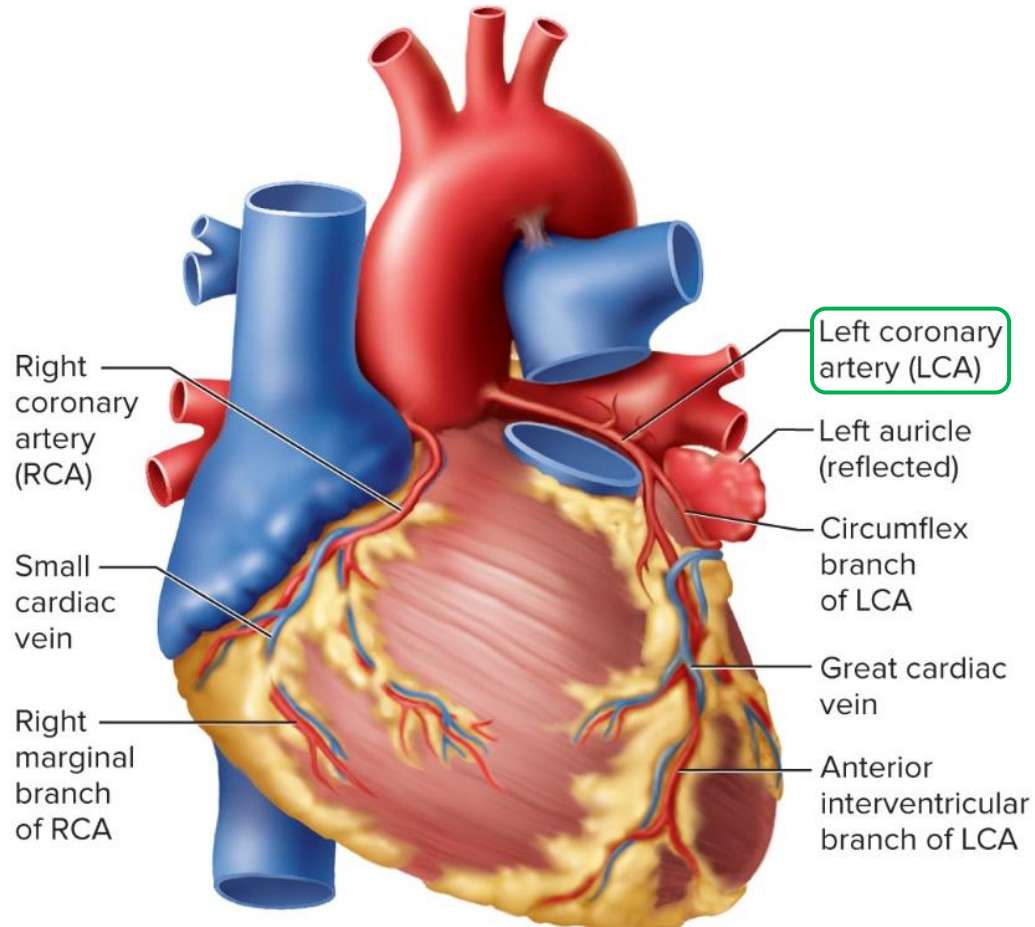


- 1 Blood enters right atrium from superior and inferior venae cavae.
- 2 Blood in right atrium flows through right AV valve into right ventricle.
- 3 Contraction of right ventricle forces pulmonary valve open.
- 4 Blood flows through pulmonary valve into pulmonary trunk.
- 5 Blood is distributed by right and left pulmonary arteries to the lungs, where it unloads CO_2 and loads O_2 .
- 6 Blood returns from lungs via pulmonary veins to left atrium.
- 7 Blood in left atrium flows through left AV valve into left ventricle.
- 8 Contraction of left ventricle (simultaneous with step 3) forces aortic valve open.
- 9 Blood flows through aortic valve into ascending aorta.
- 10 Blood in aorta is distributed to every organ in the body, where it unloads O_2 and loads CO_2 .
- 11 Blood returns to right atrium via venae cavae.

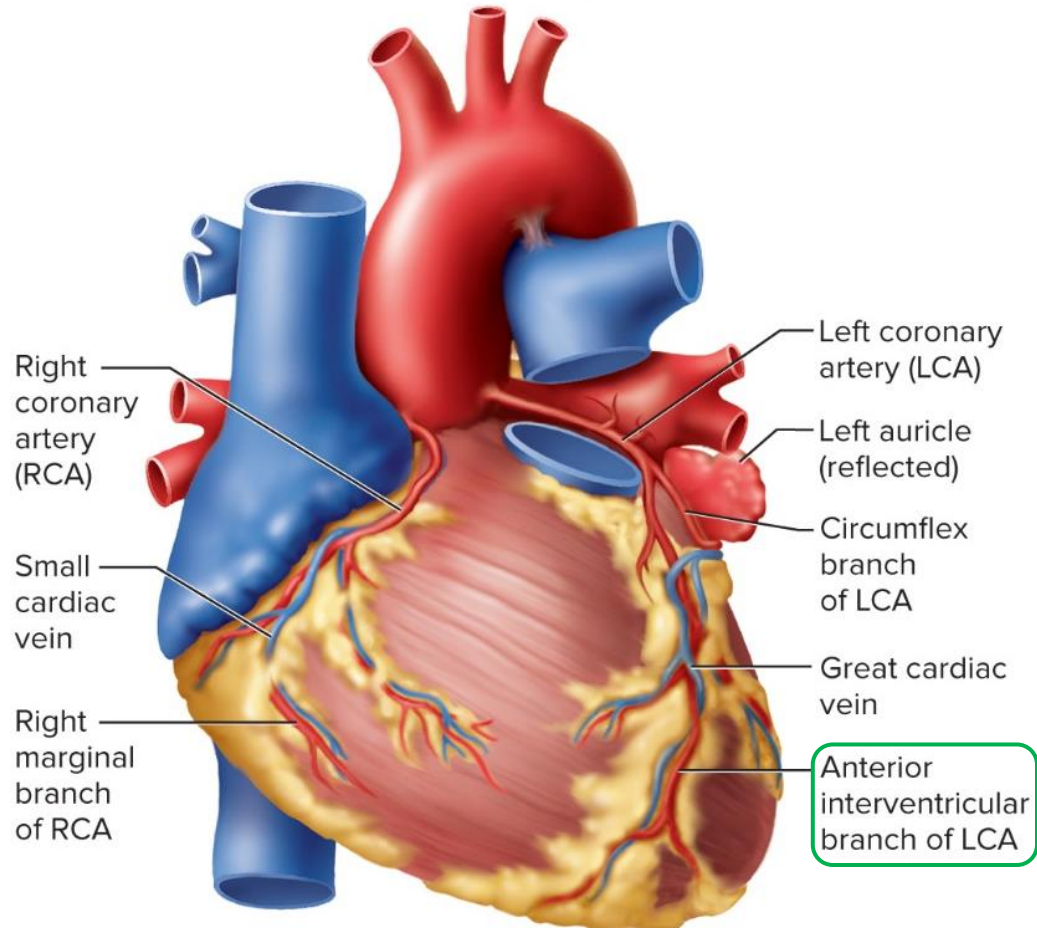
Cardiac Anatomy

*after the aorta leaves the left ventricle, it gives off a right and left coronary artery

****left coronary artery (LCA)*** travels through the coronary sulcus and divides



Cardiac Anatomy

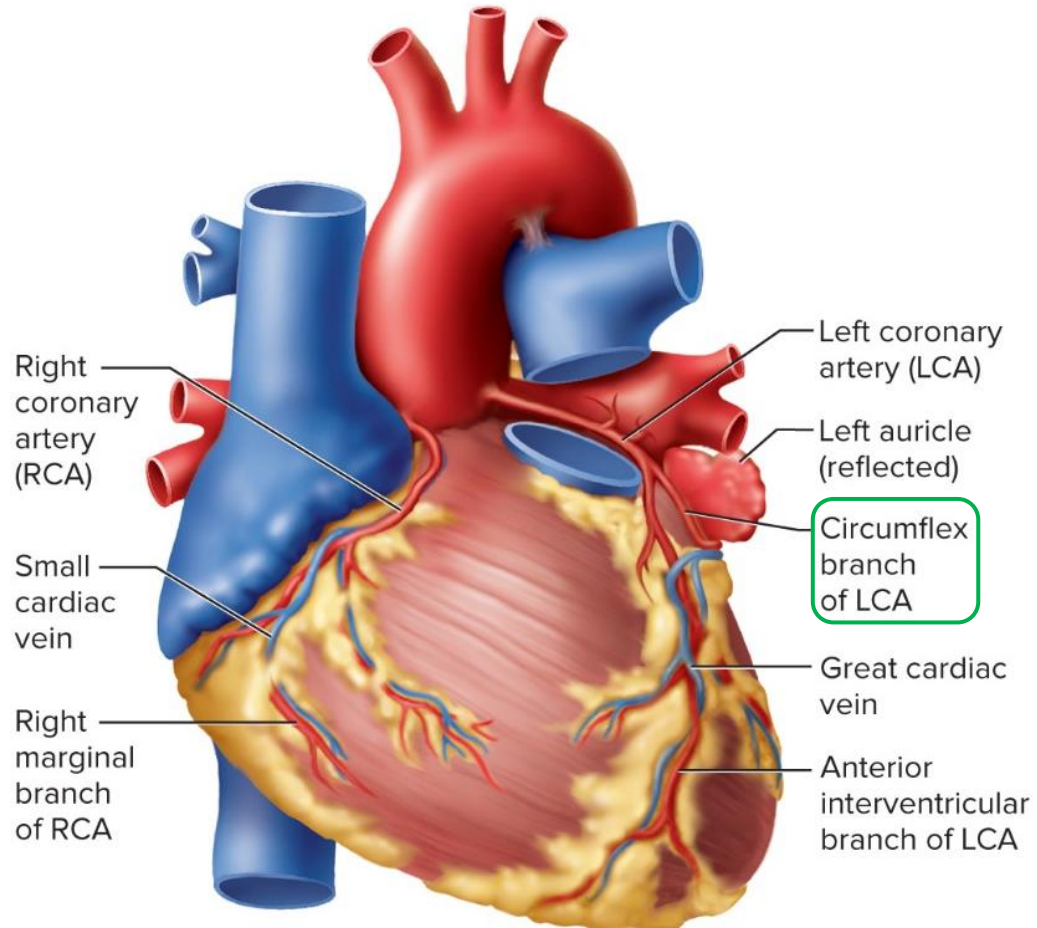


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- aka ***left anterior descending (LAD) branch***

Cardiac Anatomy



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- aka **left anterior descending (LAD) branch**

***circumflex branch** continues around the left side of the heart in the coronary sulcus

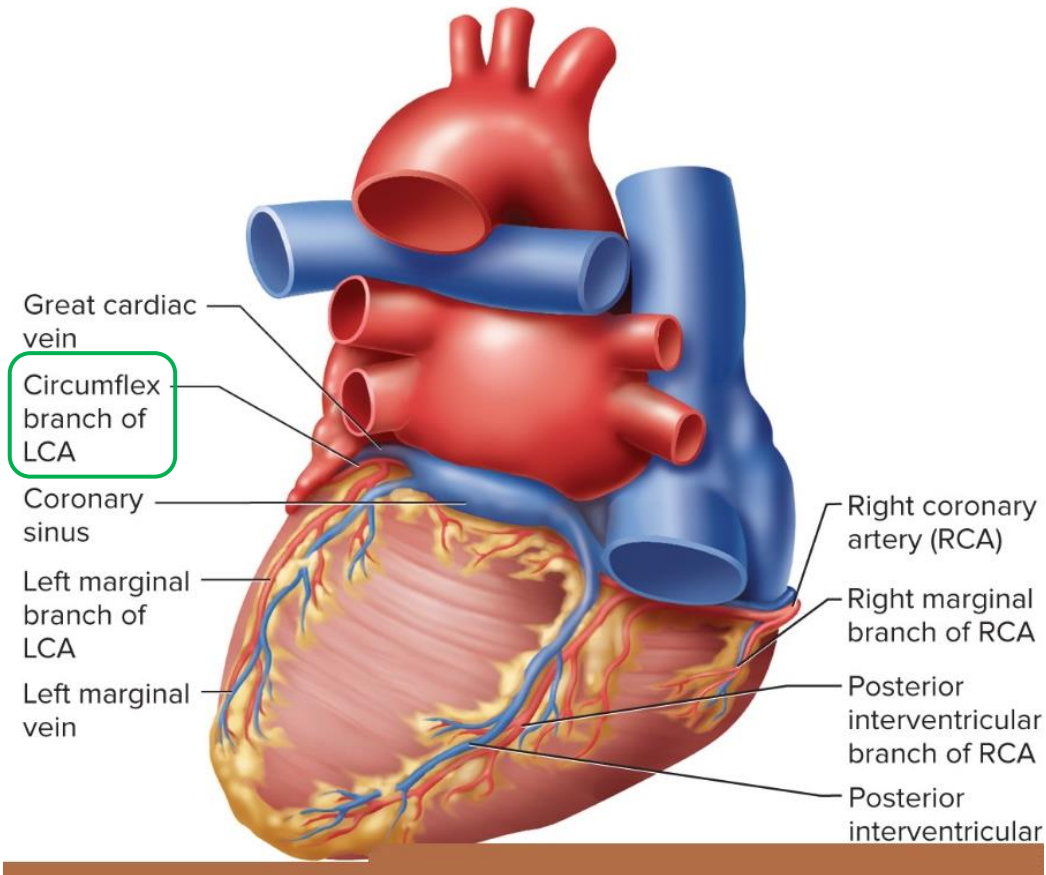
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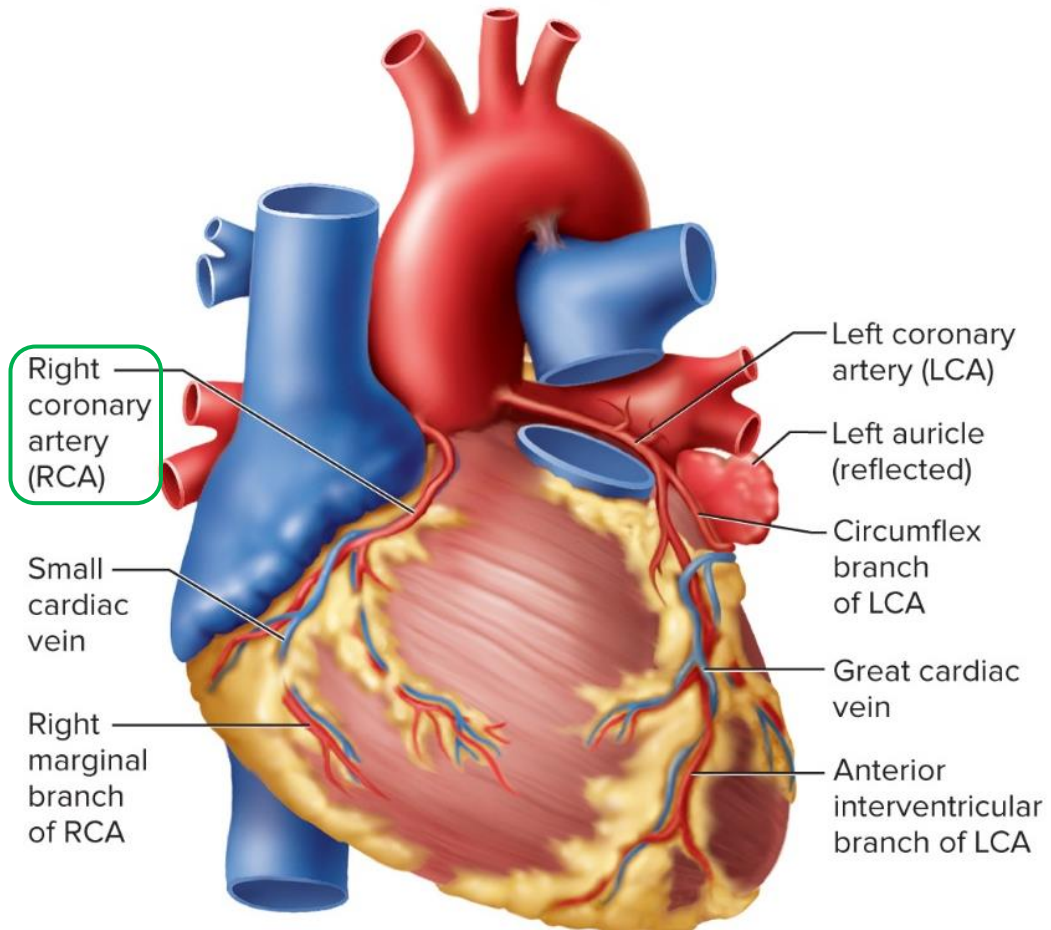
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Cardiac Anatomy

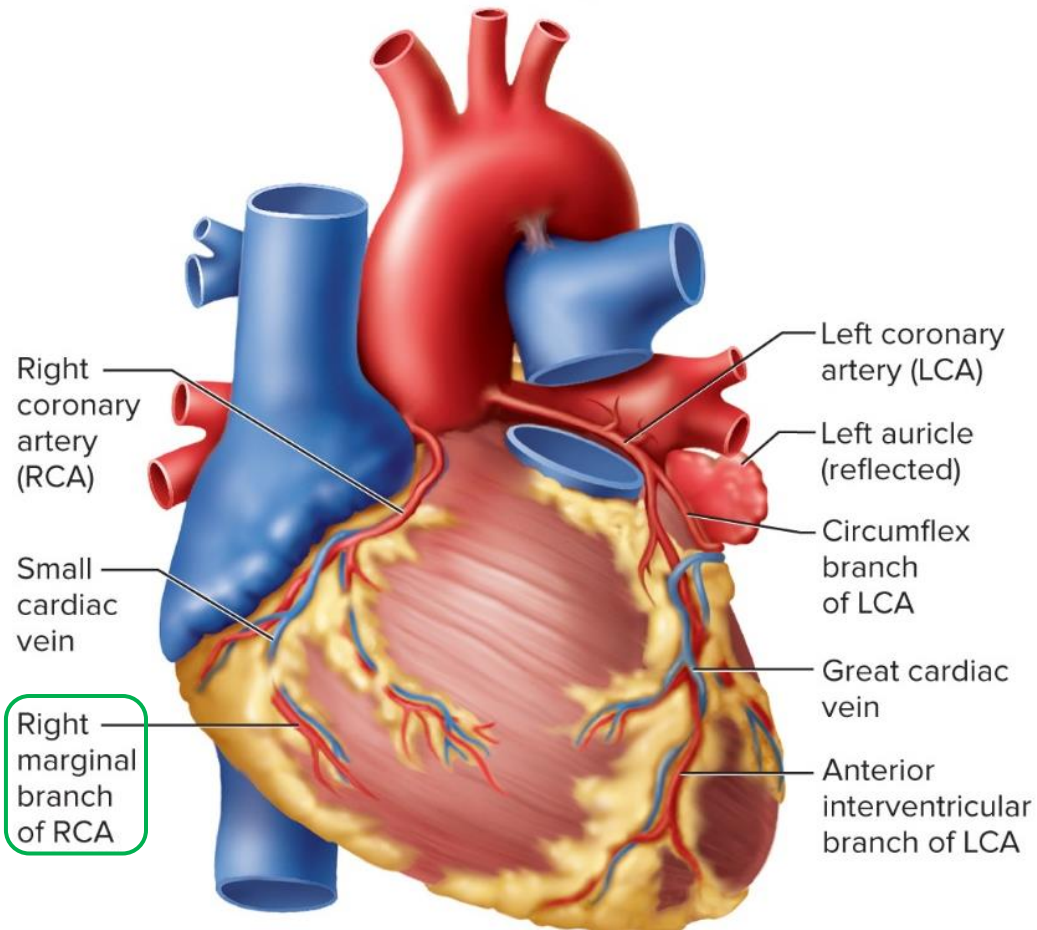
****right coronary artery (RCA)*** travels through the coronary sulcus and divides



Cardiac Anatomy

****right coronary artery (RCA)*** travels through the coronary sulcus and divides

****right marginal branch*** runs toward the apex

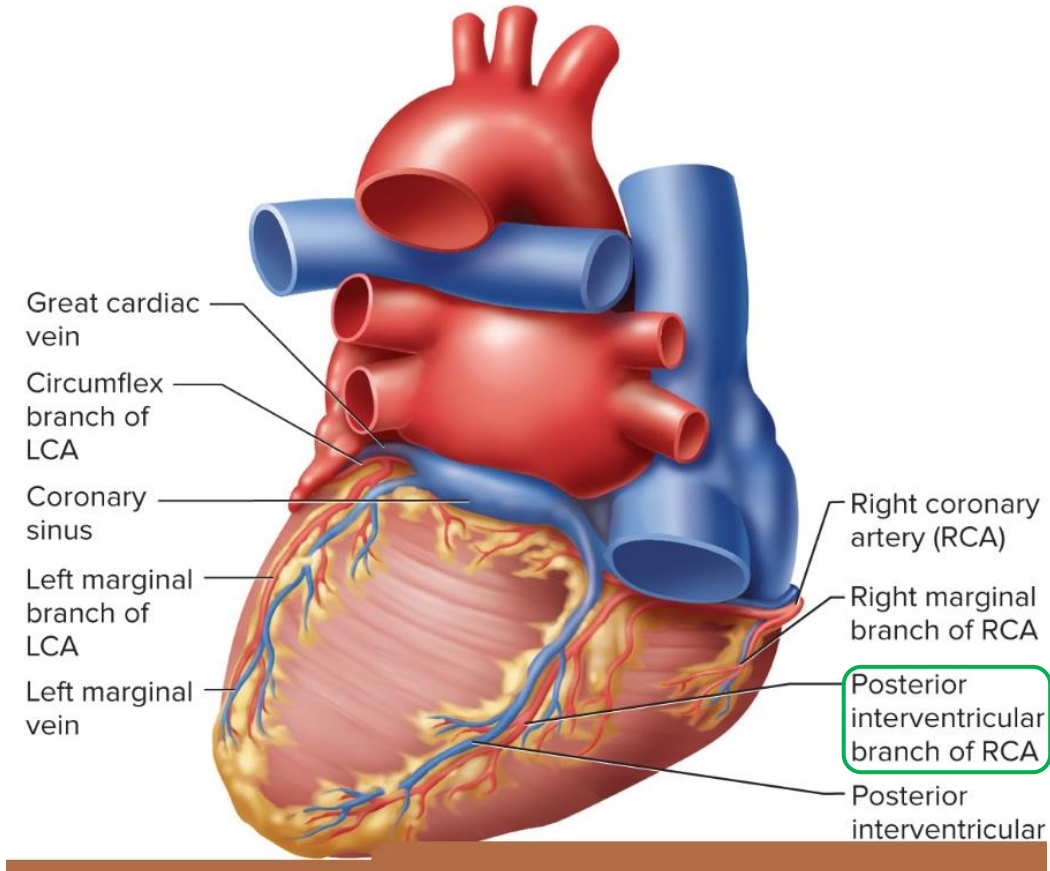


Cardiac Anatomy

****right coronary artery (RCA)*** travels through the coronary sulcus and divides

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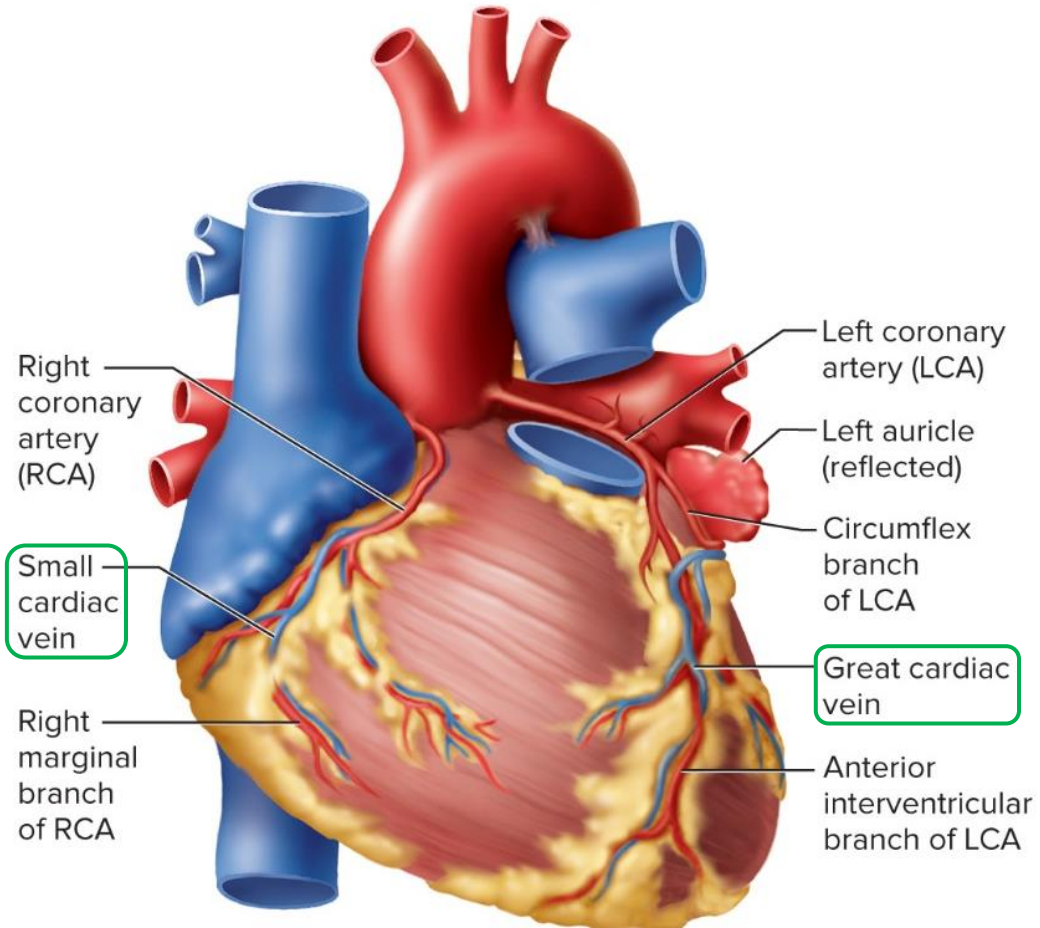
****posterior interventricular branch*** travels down the posterior interventricular sulcus



Cardiac Anatomy

****great cardiac vein*** collects blood from the anterior aspect of the heart and travels along the anterior interventricular artery

****small cardiac vein*** travels up the left side



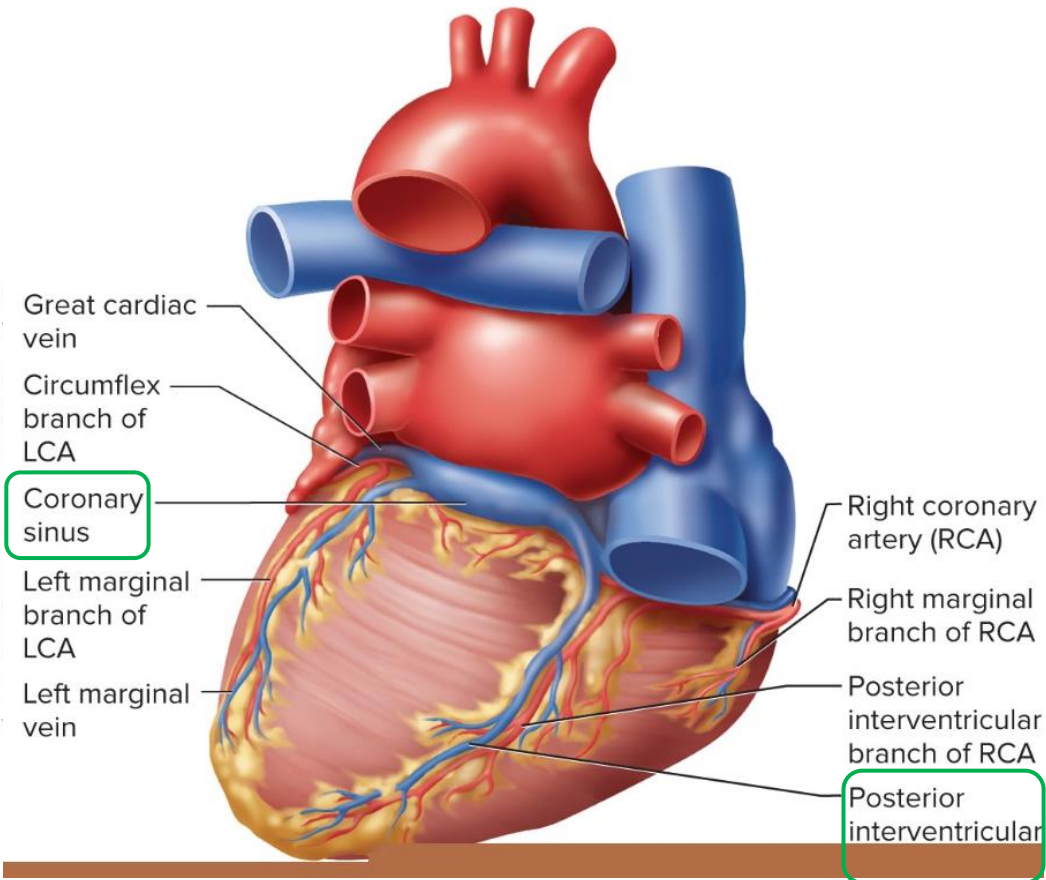
Cardiac Anatomy

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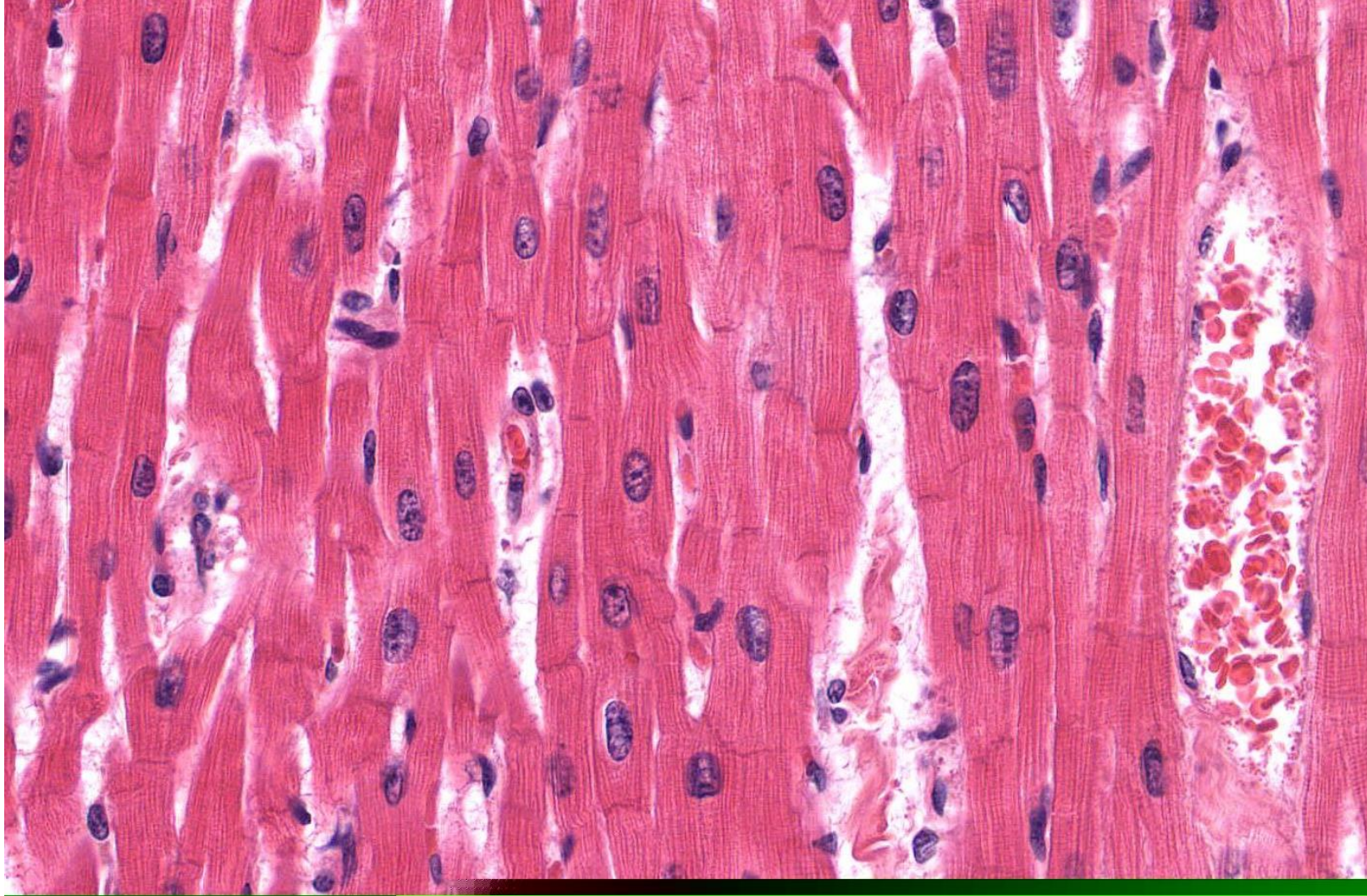
* **small cardiac vein** travels up the left side

* **middle cardiac (*posterior interventricular*) vein** collects blood from the posterior heart

* **coronary sinus** is in the coronary sulcus on the posterior side of the heart (collects blood from all veins and empties into right atrium)



Cardiac Histology



Heart Dissection

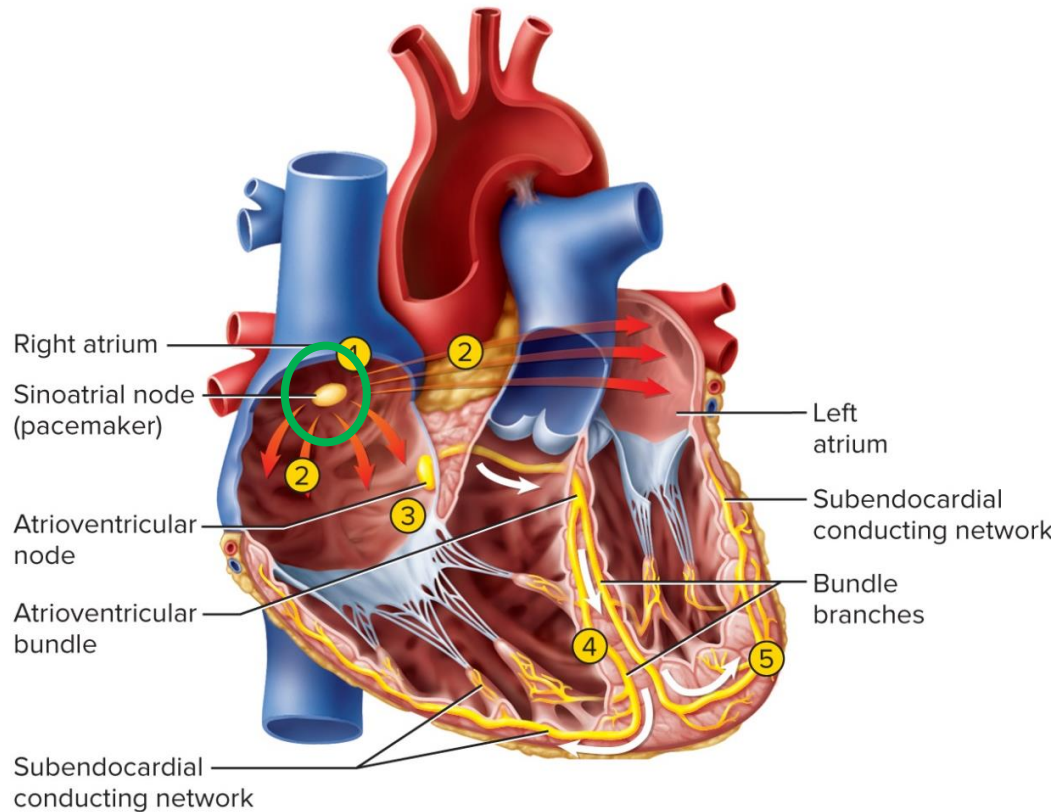


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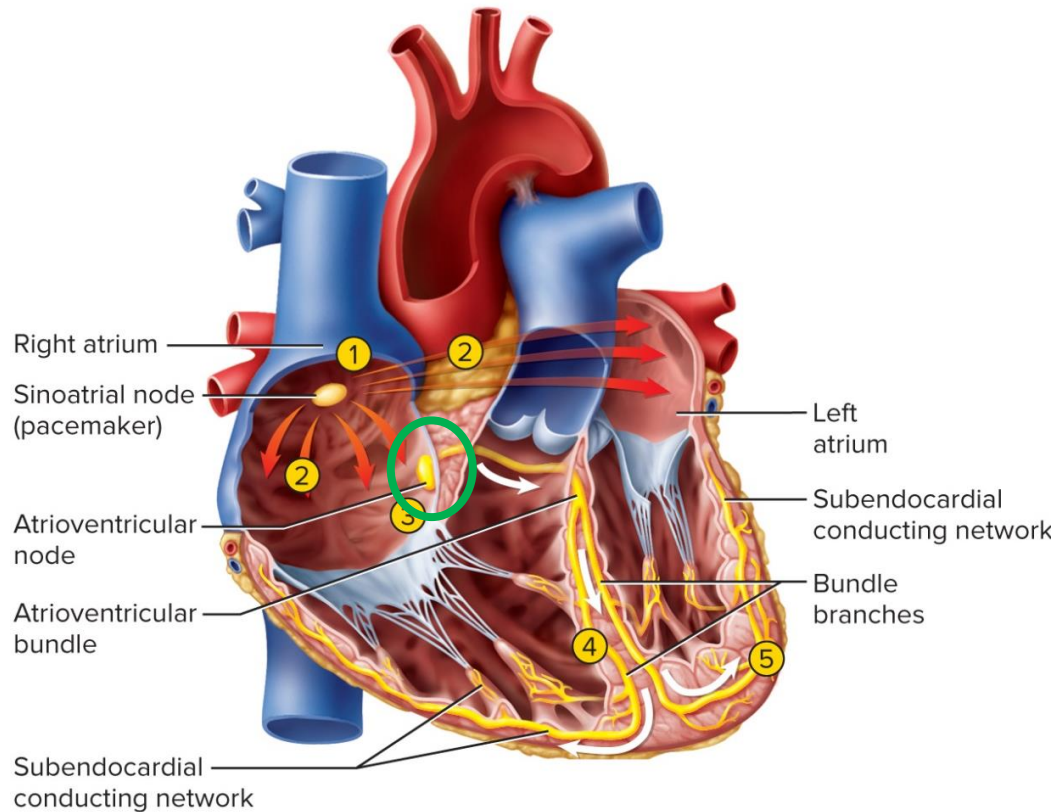
Cardiac Conduction

**sinoatrial (SA) node* is a patch of modified cardiomyocytes in the right atrium

*SA node is the *pacemaker* that initiates each heartbeat and determines the heart rate



Cardiac Conduction

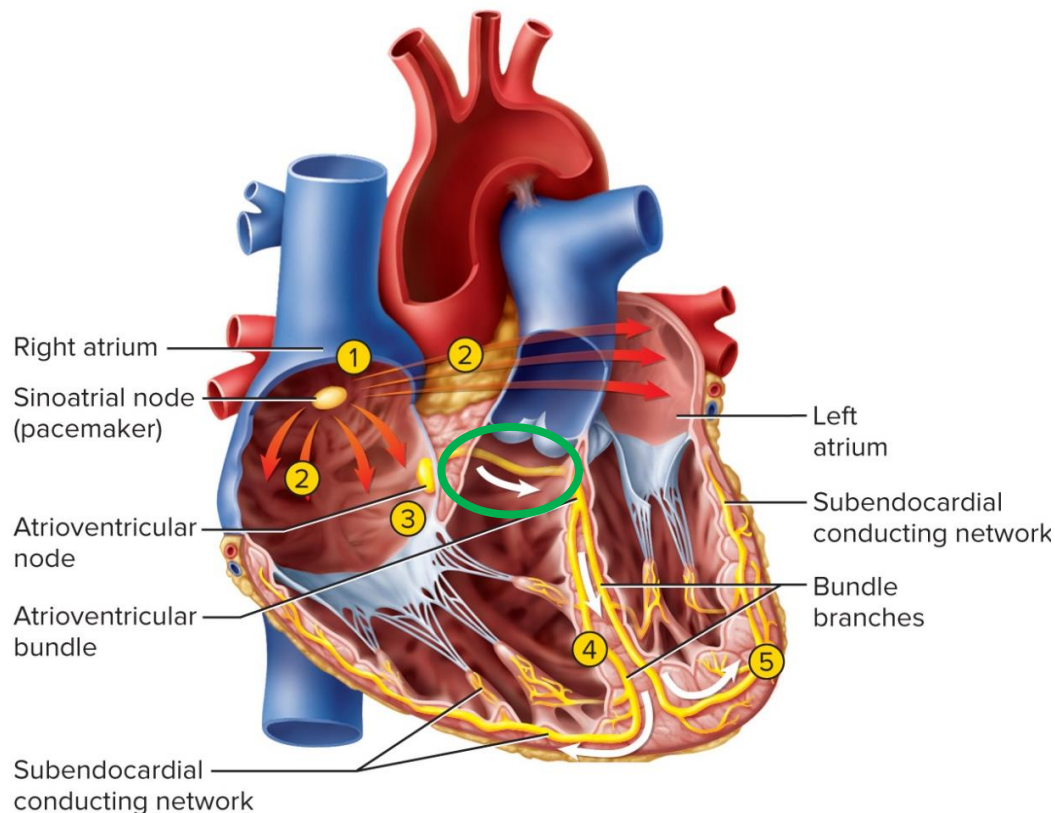


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Cardiac Conduction



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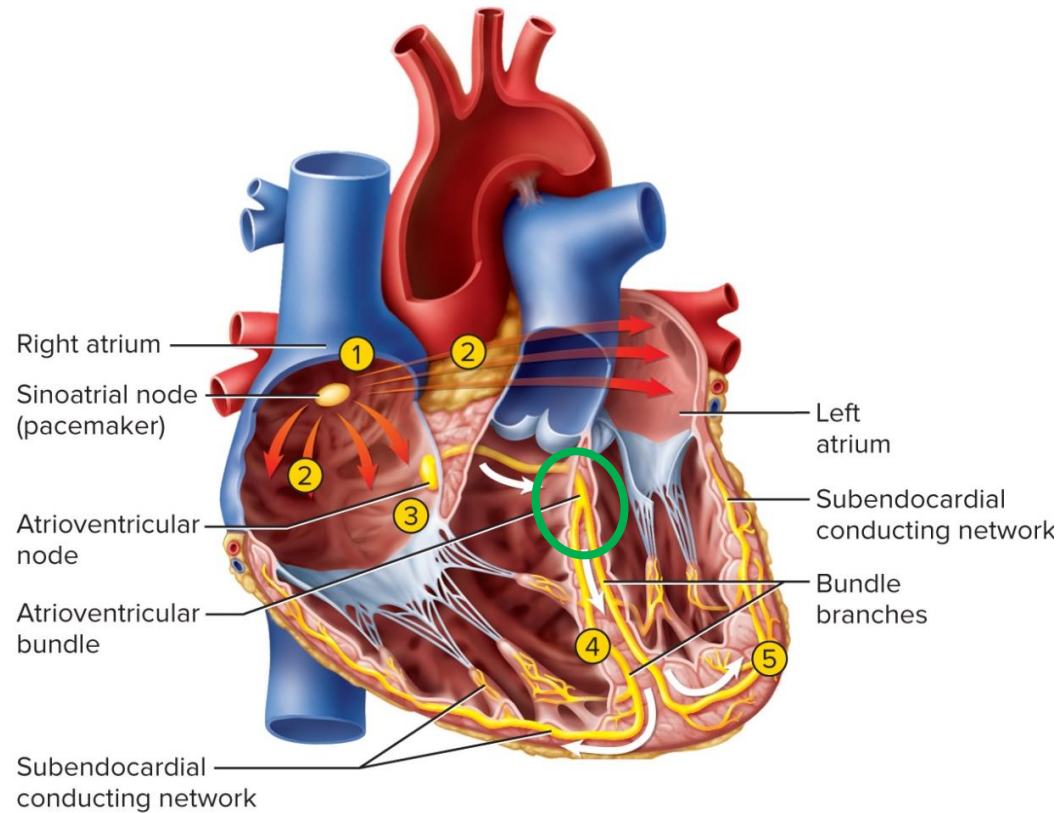
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*the ***atrioventricular (AV) bundle*** is the pathway by which signals leave the AV node

Cardiac Conduction

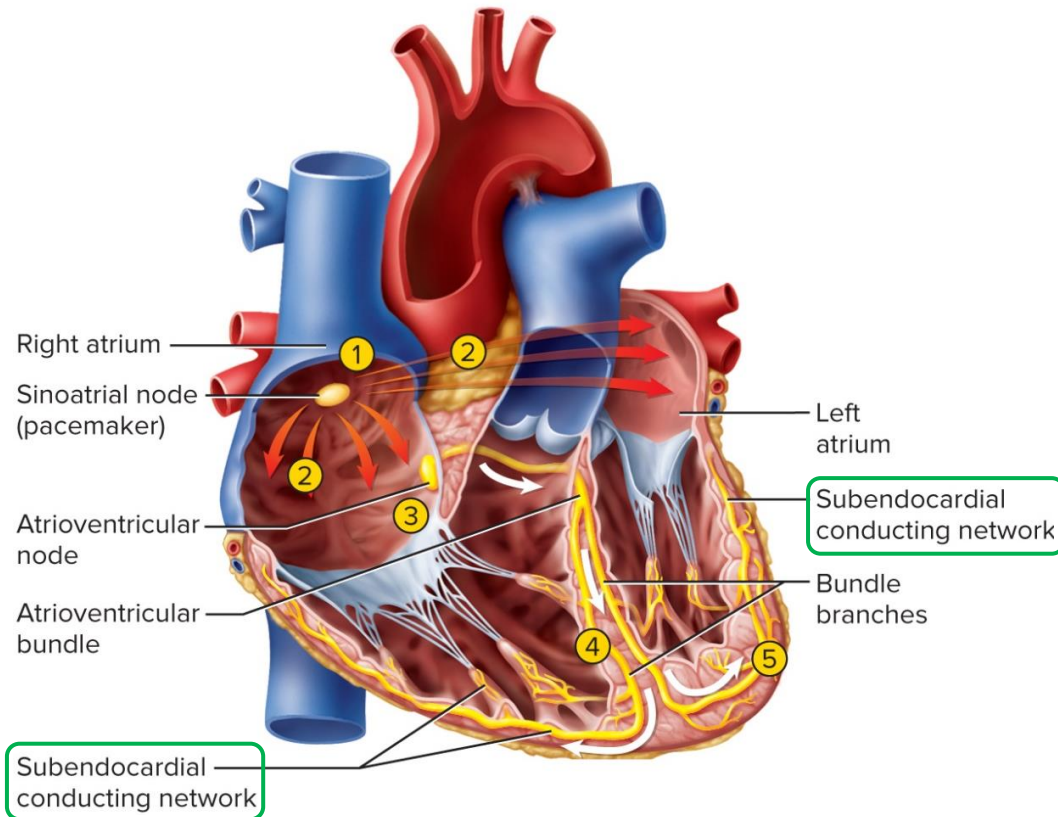
*AV bundle forks into ***right*** and ***left bundle branches***, which enter the interventricular septum and descend toward the apex



Cardiac Conduction

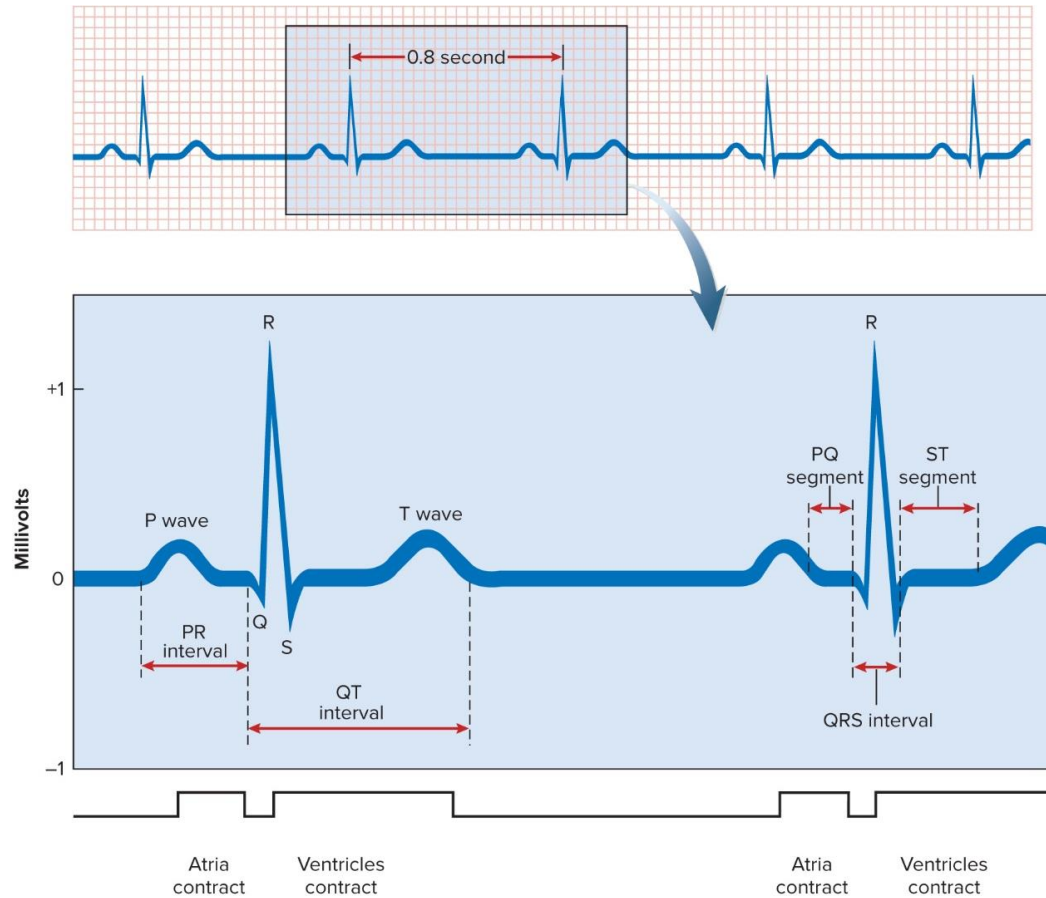
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*the ***subendocardial network*** consists of processes that arise from the lower end of the bundle branches



Electrocardiogram

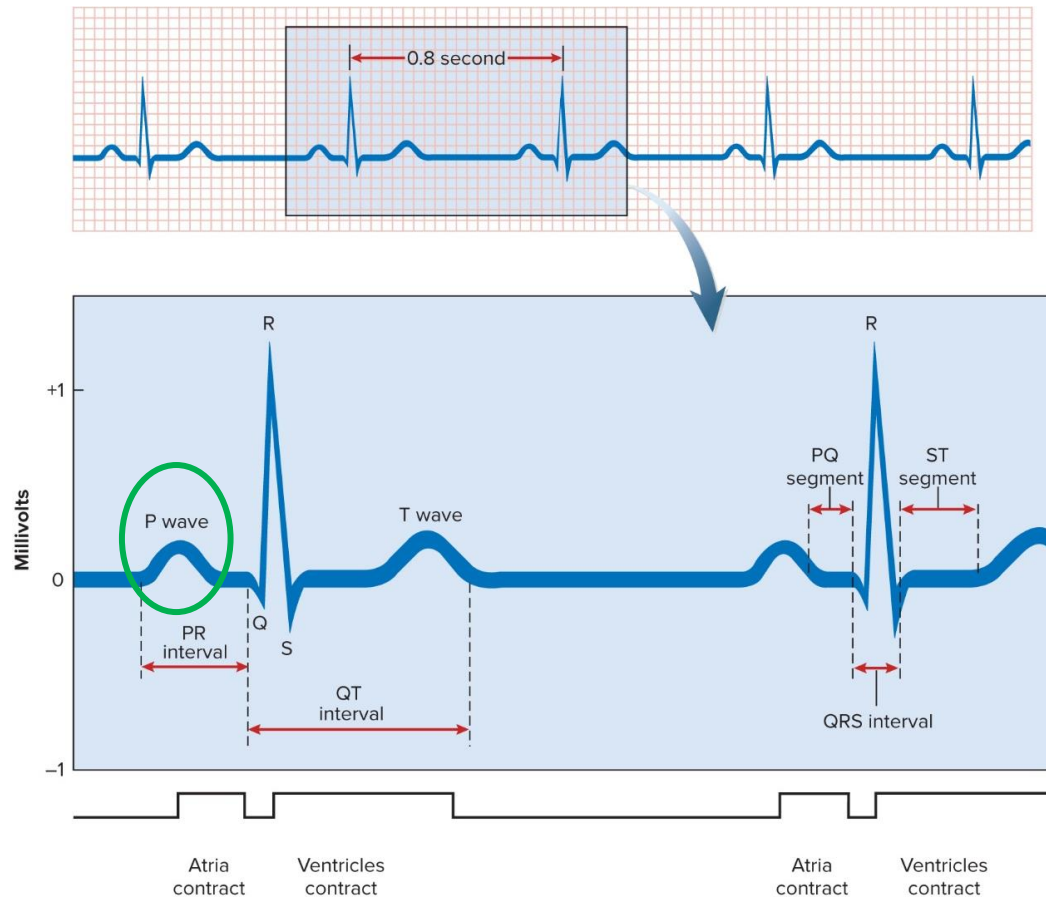
*an ***electrocardiogram*** (***ECG*** or ***EKG***) measures the electrical signals in the heart



Electrocardiogram

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*the **P wave** is produced when a signal from the SA node spreads through the atria and depolarizes them (.25 mV and .12 s)

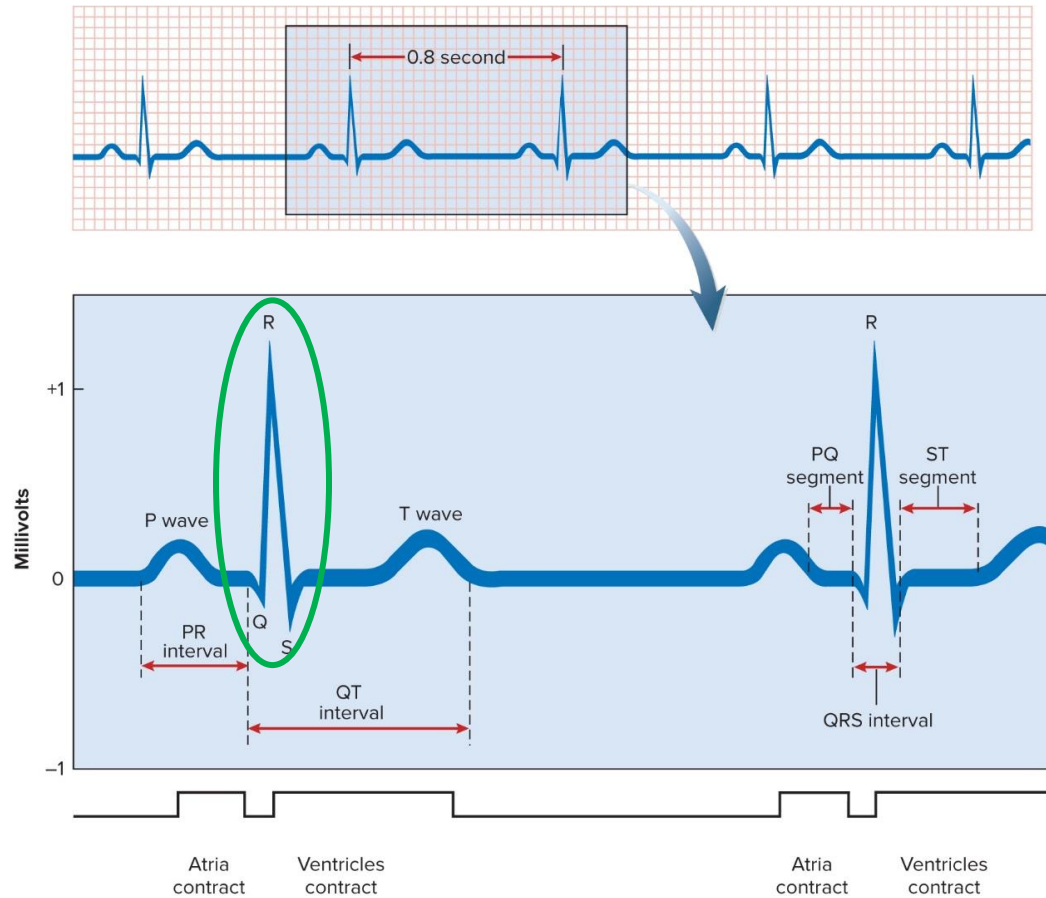


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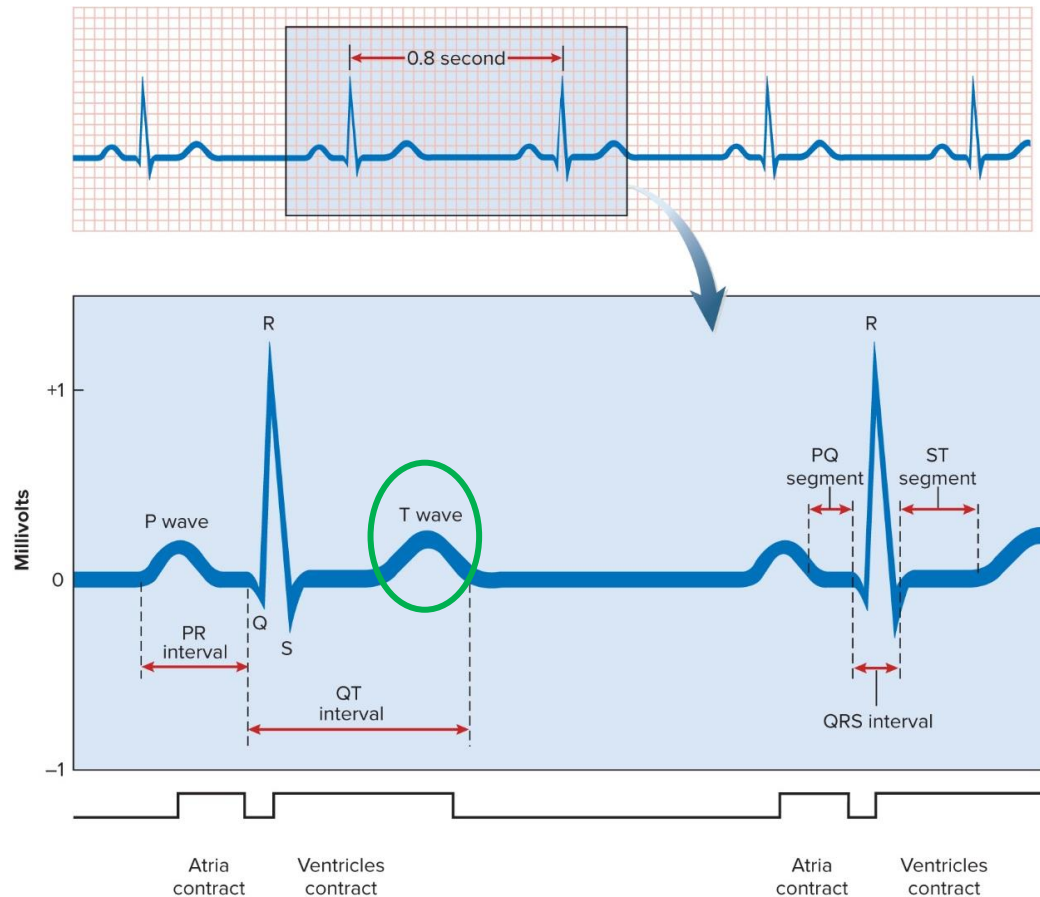
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*the **T wave** is generated by ventricular repolarization immediately before diastole



Electrocardiogram

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****normal sinus rhythm*** is 60-100 bpm

****tachycardia*** is >100 bpm

****bradycardia*** is <60 bpm